

Face Mask and Social Distance Detection Using Deep Learning Models



N. Anusha, Saumya Gupta, Y. Nikitha Naidu, M. Ruchitha,
and Richal Pandey

Abstract Coronavirus disease 2019, also known as COVID-19, has become a global threat. In dealing with this global problem of the COVID-19 pandemic, flattening the curve of COVID-19 cases will be difficult unless residents take steps to stem the spread of this virus. The most critical measures during this epidemic are wearing a mask and maintaining social distance in crowded places. This research aims to propose the efficient deep learning model among the few of the most used models in the literature to detect face mask and social distancing violations. The deep learning models MobileNet, You Only Look Once (YOLO), and ResNet-50 are implemented, and their results are compared to ascertain people in a frame and scan for social distance by calculating the Euclidean/Manhattan distance between the bounding boxes generated. Similarly, the proposed model determines if the two people are within 6 feet of the frame; if not, a warning is issued. In this research, Face Mask Lite, Real-World Masked Face, MaskedFace-Net, and Face Mask Detection datasets are considered with 90:10, 80:20, 70:30, and 60:40 ratios to train and test these deep learning models. These deep learning models are processed in Anaconda + Python 3.x (3.8 or older) environment. The proposed model can be useful in hospitals and clinics, airports, shops and workplaces, and other crowded places where social distancing and face masks are required to avert the spread of not only COVID-19 but also other communicable diseases as well.

Keywords Coronavirus disease 2019 · Deep learning models · Euclidean distance · Face mask detection · Social distance detection

N. Anusha · S. Gupta (✉) · Y. Nikitha Naidu · M. Ruchitha · R. Pandey
Department of Computer Science and Engineering, Vidya Jyothi Institute of Technology,
Hyderabad, Telangana, India
e-mail: saumya.1126@gmail.com

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023
S. Smys et al. (eds.), *Computational Vision and Bio-Inspired Computing*,
Advances in Intelligent Systems and Computing 1439,
https://doi.org/10.1007/978-981-19-9819-5_34

461

1 Introduction

According to the World Health Organization (WHO) reports, COVID-19 has affected millions of people and taken many lives. The pandemic, primarily caused by COVID-19, has sent shock waves around the world and altered people's lifestyles. The virus was first identified in Wuhan, China, in December 2019. It spread to 216 countries and seriously threatened human life. According to recent studies, the WHO declared that there have been 585,950,085 cumulative COVID-19 cases and 6,425,422 cumulative deaths all over the globe with the number of cases in India accounting for 44,223,557 [3]. Many people have died during this time, and others are in dangerous situations. Almost every country has experienced the consequences of lockdown and the loss of loved ones. Symptoms of COVID-19 include cough, fever, loss of taste (ageusia), loss of smell (anosmia), and less common symptoms such as headache, nasal congestion, runny nose, muscle aches, diarrhea, sore throat, eye irritation. The virus spreads easily when people breathe air contaminated with virus-containing droplets/aerosols and small airborne particles [4]. The government has issued guidelines such as wearing face masks, maintaining social distance, and vaccination to halt the spread of COVID-19 and other deadly viruses.

Adhering to precautionary measures to avoid COVID-19 is essential in places where people gather in large numbers, such as shopping malls, hospitals, airports, and workplaces [1]. In order to make people aware of the conditions, emerging technologies should be developed that capture video (live or recorded) or pictures of people in a public place and also detect whether everyone conforms to the norms of avoiding precautions. It should also consider the side effects and inform people about them to maintain social distance and wear a mask with an alarm or voice message would be beneficial. This would encourage people to be more careful to protect themselves and those around them from this disease.

2 Literature Survey

To understand the severity of the need for this research and to get a better idea of the deep learning models and datasets that can be used, research has been carried out on previous published works in the literature, and a clear note has been made as given in Table 1.

From Table 1, it can be observed that YOLOv3 and YOLOv5 [1, 5–7, 9, 10, 14, 16, 17], ResNet-50 [1, 7], and MobileNet [8, 14, 19] are the models which are widely used. Table 1 describes 15 research papers with respect to face mask and social distance detection, applied models, distance measurement techniques, and datasets used. The COCO [9], WIDER FACE [5], Real-Time Masked Face Datasets (RMFDs) [7, 14], and Google images are widely used datasets.

From the survey, it can be observed that most of the work has used Euclidean distance to calculate social distance. Some common disadvantages are the lack of

Table 1 Related work

Author(s) and year published	Description	Applied models	Distance measurement techniques applied	Dataset
Bhambani et al. [5]	YOLOv4 object detection model is used on real-time images and video footage	YOLO version 4 for face mask and social distance detection	Focal length and sensor dimensions to measure the distance between two individuals	7959 images from MAFA and WIDER FACE datasets
Singh et al. [6]	Model performance: 91.2% accuracy, 90.79% F1-score, and 7.12 s average prediction time for 78 frames of video	Person detection: YOLOv3, DBSCAN Face mask detection: Dual Shot Face Detector (DSFD) and MobileNetV2	Euclidean/Manhattan distances	1792 photos have been classified with zero for no masks and one for masks
Walia et al. [7]	Model results: Accuracy: 96% using Stacked ResNet-50 (better performance than other algorithms)	Human Detection: YOLOv5 Face Mask Detection: Stacked ResNet-50 Proximities within the persons: DBSCAN	Euclidean distance	Real-Time Masked Face Dataset (RMFD), Google images
Limbasiya and Raut [2]	Precision score: 91.7% Confidence score: 0.7	ResNet-50 and YOLOv3 are combined with a transfer learning technique to achieve a balance of resource constraints and recognition accuracy	Euclidean distance	3165 custom images with different types of masks
Krishna et al. [8]	The framework was designed based on deep learning techniques for detection	Method used: MobileNet Training and testing: CNN model	Manhattan distance	Custom dataset from different sources

(continued)

Table 1 (continued)

Author(s) and year published	Description	Applied models	Distance measurement techniques applied	Dataset
Hou et al. [9]	Violation of face mask and social distance is indicated by a red colored box frame and a red color line	YOLOv3 is used for person detection OpenCV is used for converting the camera view into a 2D view by selecting four points. Euclidean is used to find the distance in 2D view	Euclidean distance	COCO dataset
More et al. [10]	Video → split into images → preprocessing → Segmentation → Feature Extraction → CNN and YOLO	YOLOv3 for detecting persons CNN algorithm for detecting face masks	–	Custom Personal Images and Custom Video Footage Dataset (CVFD) dataset
Hassan et al. [11]	Accuracy: 91.7% Confidence: 0.7 A memory value of 0.91 with FPS = 28.07	MobileNetV2 and Single Shot Object Detection (SSD) is used for object detection	–	Custom dataset of 3165 images
Sridhar et al. [12]	Gives drawbacks of YOLO and applies SSD and MobileNet algorithms Accuracy > 95%	SSD for Object Detection OpenCV, TensorFlow, Keras, and MobileNetV2 for face mask detection	–	Custom dataset
Borika et al. [13]	Sound waves are used in order to calculate the distance between two human bodies using sensors. Here, variation is determined using received signal strength value (RSSI)	This model uses the SSD framework and MobileNetV2 model and is based on ROI	–	Custom dataset

(continued)

Table 1 (continued)

Author(s) and year published	Description	Applied models	Distance measurement techniques applied	Dataset
Nikhil et al. [14]	As it is combined with both MobileNetV2 architecture and Single Shot Detector (SSD) framework, this model will have a quick, efficient deep learning-based method	YOLOv3, it gets the bounding boxes The SSD framework which comes with MobileNetV2 architecture is used for mask detection	Euclidean distance	A custom dataset is created using Google/Bing search API, Kaggle datasets, and RMFD dataset with about 3835 photos
Gawde [15]	YOLOv3 seemed to be the best Age estimation is also done to understand and notify the severity of the risk prone to the people of that age	In the case of multiple boxes, YOLOv2 and faster R-CNN are used to obtain bounding boxes and identify face masks and social distancing. On 1000 images, the age classification network was refined	$d = ((2*3.14*1.8)/(w + h*360) * 1000 + 3)$, where w = width of the bounding box and h = height of the bounding box	Real-World Masked Face Dataset given by X-Zhangang consisting of 90,000 faces and Kaggle were also used
Kumar and Shetty [16]	YOLO object detection is used to locate the people in the frame and used Euclidean distance for social distancing Developed an Android application named “StaySafe”	TensorFlow and Keras to train a CNN model OpenCV was used to capture the video YOLO object detection finds Euclidean distance	Euclidean distance	A custom dataset with 3800 images, 1900 images each for “with mask” and “without a mask” categories, is used
Sathyabama et al. [17]	The YOLOv3 is used along with TensorFlow, using which each object detected is monitored within a box, and then, they are compared with each other to check for the distance between them and its accuracy is above 96%	YOLO for object detection, TensorFlow for easy portability and efficient usage of ML, deep learning and scientific computing, R-CNN, and the ResNet-50	Euclidean distance	There are 1900 images from Google images, Kaggle, and open-source image libraries, both with and without masks

(continued)

Table 1 (continued)

Author(s) and year published	Description	Applied models	Distance measurement techniques applied	Dataset
Kaviya et al. [18]	Continuous individual identification is performed by Single Shot Article Detection (SSD) along with MobileNetV2, and OpenCV, achieving an accuracy of 91.2%	The core model for recognition is MobileNetV2 engineering. The MultiBox Detector SSD is used for excellent image characterization, and the Euclidean distance formula is employed	$F = (P*D)/H(P*D)$, where H = persons' height P = pixel stature of the individual D = distance away from camera $D = (H \times F)/P(H \times F)$	A custom dataset of 3165 photos is used, containing images with mask and without a mask

recognition of blurry images and faces covered by a scarf or hands. The distance of people from the camera made social distance detection complicated.

3 Datasets Used

3.1 COCO Dataset

The Common Objects in Context (COCO) dataset is used for object detection, segmentation, and captioning. It contains 330,000 images, more than 220,000 of which are labeled. It is divided into 80 object categories and 91 stuff categories [23]. COCO supports a variety of annotation types, including key point detection, object detection, dense pose, stuff segmentation, panoptic segmentation, and image captioning. JSON is used to store the annotations [20].

3.2 WIDER FACE Dataset

Images taken from the WIDER dataset make up the face detection dataset known as WIDER FACE. It consists of 32,203 images and labels, furthermore as 393,703 faces with a large variety of scale, pose, and occlusion. The WIDER FACE dataset is split into sixty-one event classes. It arbitrarily selects 40, 10, and 50% of the data for every event class as training, validation, and testing sets. It uses the same evaluation metric as the PASCAL VOC dataset [21].

3.3 RMFD Dataset

The Real-Time Masked Face Dataset (RMFD) is a large dataset for detecting masked faces. There are 24,771 items, two classes, and 24,771 labels. After cleaning and labeling, it has 90,000 recognizable faces and 5000 faces with masks of 525 individuals. By masking the faces in publicly available face datasets, a collection of 500,000 faces from 10,000 subjects known as the simulated masked face recognition dataset was created. The datasets created are used to create and train a face-eye-based multi-granularity masked face recognition model [22].

4 Methodology

To better understand the system of applied models, a clear description of all the algorithms used in this research is discussed in this section. Three models, namely, ResNet-50, YOLOv5, and MobileNet have been implemented to identify the persons, to detect the face mask and social distancing between the persons. The overall process applied is depicted in Fig. 1.

4.1 ResNet-50

Considerable study has been undertaken to identify facial masks and monitor breaches of social distance, but none has succeeded in coming up with an integrated system for both. Here, an integrated approach to detect face masks in humans and monitor social distancing violations. The recognition of masks and the anomalies in social distancing norms are two major concerns in containing the spread of COVID-19. The proposed research is conducted in two parts—first, face mask classification using ResNet-50 and social distancing monitoring using DFDS clustering. The first step is to extract a single frame from a CCTV video. It is then processed with YOLOv3 to detect people and draw bounding boxes on each detection. These inputs are then used for facial detection. The detected faces are then given as input to ResNet-50 for face mask detection and classified as masked (green color) or unmasked (red color). Figure 2 shows the pipeline diagram of the methodology applied for ResNet-50.

One of the ResNet versions is ResNet-50, a convolutional neural network with 50 layers. It has one Average Pool layer, one MaxPool layer, and 48 Convolution layers. Deep residual learning serves as the foundation for ResNet. It resolves the vanishing gradient issue even with very deep neural networks. ResNet-50 is significantly smaller than previous designs while having 50 layers and more than 23 million trainable parameters. Its performance is still under discussion, but understanding it is made clearest by describing residual blocks and how they work.

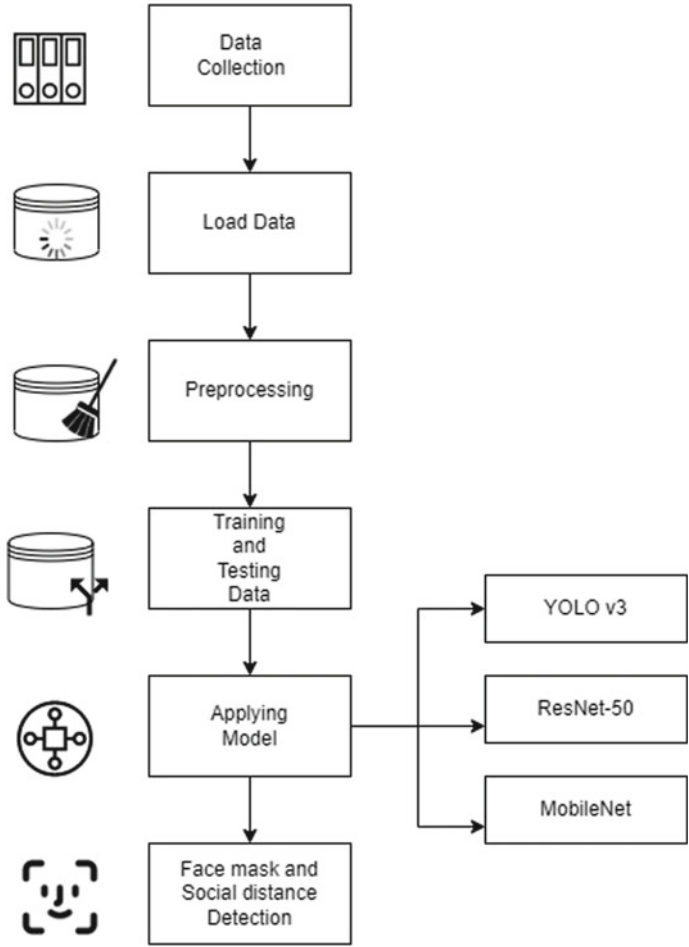


Fig. 1 Overall process applied

4.2 YOLO

YOLO stands for “You Only Look Once.” It is a family of object detection models introduced in 2016 by Joseph Redmon as “You Only Look Once: Unified, Real-Time Object Detection.” Others later improved on it for better understanding and performance, reliability, and accuracy.

A combination of custom data and datasets from Kaggle, a total of 3800 images, 853 images from three classes, and the PASCAL VOC format is adopted for their bounding boxes. The classes are “with mask,” “without a mask,” and “mask is worn incorrectly.”

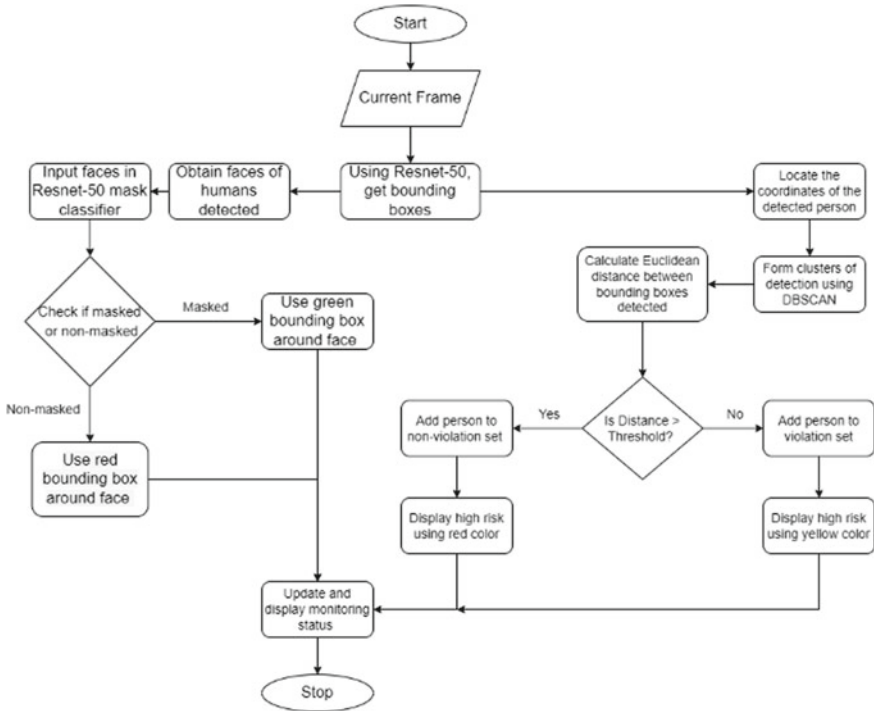


Fig. 2 ResNet-50 pipeline diagram

YOLOv5 is used for social distancing, and the distance is calculated using the mathematical formulae mentioned in Eq. 1. YOLOv5 pipeline diagram is shown in Fig. 3. Green boxes are generated around people who follow the social distancing, and red bounding boxes are generated around people who do not follow the social distancing.

$$\text{distance}(a, b) = \sqrt{\sum_1^n (b_i - a_i)^2} \quad (1)$$

where

a, b two points in Euclidean space

a_i, b_i Euclidean vectors

n n-space.

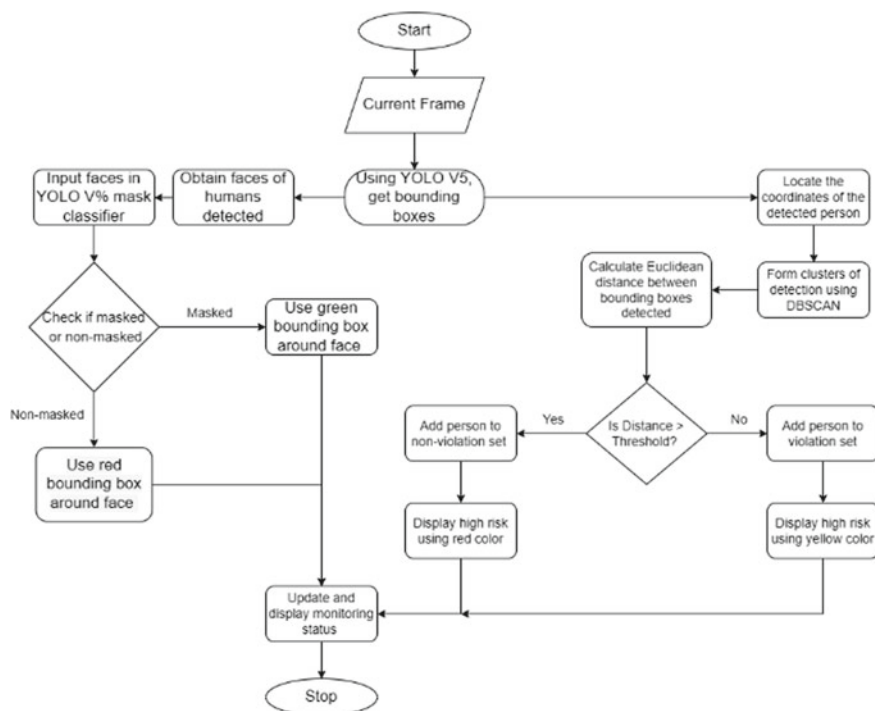


Fig. 3 YOLOv5 pipeline diagram

4.3 MobileNet

It begins by collecting data from various people, either with or without mask images. The trained model can differentiate between people wearing masks and those who do not. The proposed method makes use of a dataset of 3853 images. A batch size of 32 and epochs of 10 are taken for this.

MobileNet architecture. The essential layers of MobileNet, which become depthwise separable filters, are described in this section. The core of the MobileNet model is the depthwise separable convolutions, a type of factorized convolutional that separates a conventional convolution into a depthwise convolution and a pointwise convolution. Thanks to depthwise convolution, each input channel in MobileNet receives a single filter. The pointwise convolution is used to combine its outputs using a 1×1 convolution. Both filters are obtained in addition to the outcome in one step using a typical convolution. The depthwise separable convolution separates this into two layers: a layer for combining and a layer for filtering. The computing and model size are significantly decreased as a result of this factorization. Figure 4 demonstrates the resolving of a typical convolution (a) into a depthwise convolution (b) and a 1×1 pointwise convolution (c). A standard convolutional layer receives as input a DF

$\times DF \times M$ feature map F and outputs a $DF \times DF \times N$ feature map G , where DF appears to be the spatial breadth and height of a square input feature map 1. There are N output channels, DG is the spatial height and breadth of a square output feature map, and M is the number of channels (input depth and output depth). Convolution kernel K is of size $DK \times DK \times M \times N$, where DK is the spatial dimension of the conventional convolutional layer [1].

Preprocessing, training, and testing. Preprocessing is phased in conjunction with train–test splitting. The four steps involved are to convert an image to an array, resize the data, label binarizer, and image data generator. Picture scaling is an essential preprocessing step in object recognition due to the efficacy of learning algorithms. If the image is smaller in size, the model will score higher. In this analysis, scaling the image results is divided into 224×224 pixels. The subsequent step involves creating an array from all of the images in the dataset. The image is regenerated into an array so that the repeat function can access it. The image will then be used as preprocessed data by MobileNetV2.

Equations 2, 3, 4 and 5 are used to calculate precision, recall, accuracy, and F1-score, respectively [19]. The overall process flow is shown in Fig. 5.

$$\text{Precision} = \text{tp} / \text{tp} + \text{fp} \quad (2)$$

$$\text{Recall} = \text{tp} / \text{tp} + \text{fn} \quad (3)$$

$$\text{Accuracy} = \text{tp} + \text{tn} / \text{tp} + \text{tn} + \text{fp} + \text{fn} \quad (4)$$

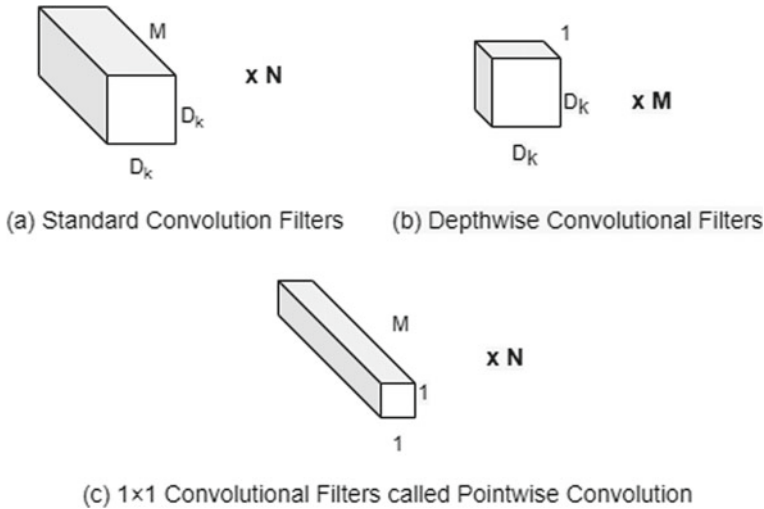


Fig. 4 Factorization of typical convolution. *Source* Ahama et al. [1]

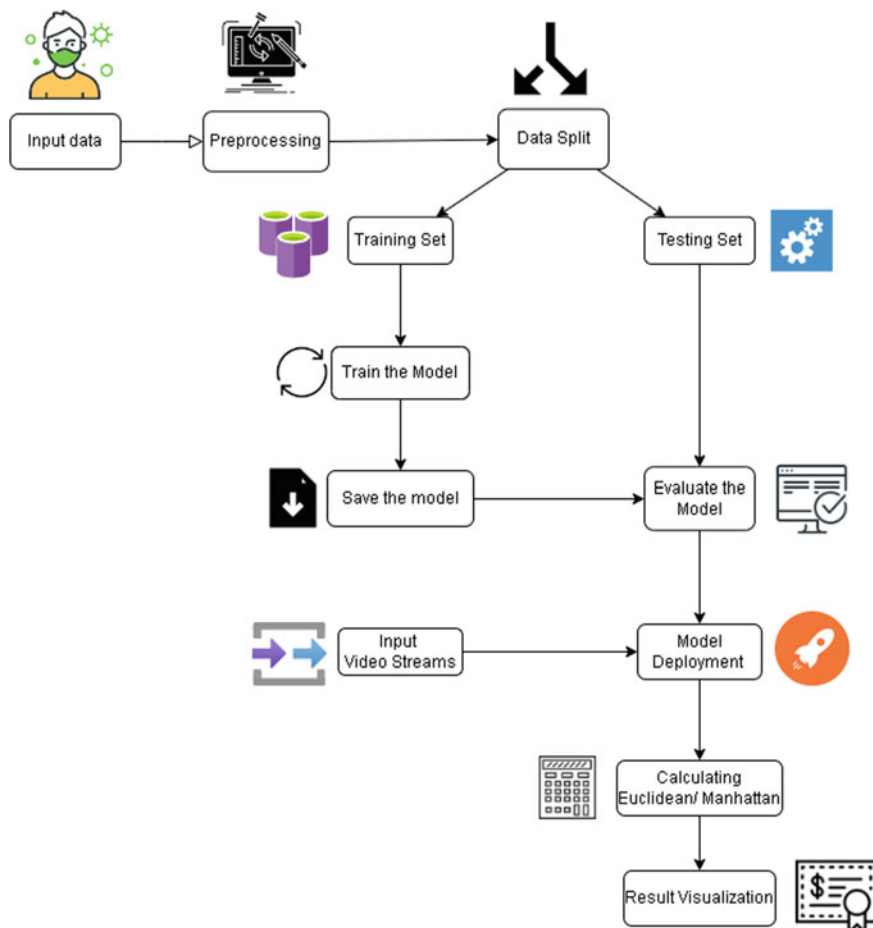


Fig. 5 Overall flowchart of the face mask and social distance detection models

$$F1\text{-score} = 2 * (\text{precision} * \text{recall} / (\text{precision} + \text{recall})) \quad (5)$$

4.4 Confusion Matrix

Confusion matrix is used in this research, to evaluate how well the classification models perform for a certain set of test data. The confusion matrix helps to find how well a classification model (also known as a “classifier”) performs on a set of test data for which the true values are known.

The dataset's real labels and the model's predicted values are mapped out in a matrix. The number of entries in each class using the matrix can be determined. These classes are:

- True negative (tn): The model's forecast was No, and the actual or real value was also No.
- True positive (tp): Both the actual value and the model's prediction were true.
- False negative (fn): No was predicted by the model, but the actual value was Yes, and it also goes by the name Type-II mistake.
- False positive (fp): Although the model projected Yes, the actual result was No. Another name for it is a Type-I mistake.

5 Results and Discussions

The implemented models on the input dataset (refer Fig. 1) in this research aids in carrying out the social distance evaluation procedure automatically. Frame in the considered input video is shown in Fig. 6. Following training with the custom dataset and pre-trained weights, the bounding box with the labels and the confidence score are displayed above the box to demonstrate the model's accuracy on the test dataset (COCO). These models first recognize every person in the cameras' range of view and use a green color bounding box around to represent persons who are distant from others and with a red color bounding box to represent persons who are not (refer Figs. 7, 8, 9, 10, 12, 13, 14 and 15). Secondly, the model detects and represents the people maintaining the social distance with yellow color line and red color between the people who are not following social distancing rules (refer Fig. 11). These models also detect and recognize a person with the blue color bounding box around their face if they are not wearing a mask and with a green color bounding box if they are wearing a mask (refer Figs. 7, 8, 9, 10, 12, 13, 14 and 15).

This input video (refer Fig. 6) is used to test the models stated in this research to determine how well and accurately face masks and social distancing are detected.

Figure 7 shows the results of person and face mask detection applied on a frame of video from a security camera. The ResNet-50 model analyzes video frame by frame, identifying every person inside the cameras' field of view. Figure 6 also shows the total people count and also displays the number of safe and unsafe people with green colored and red colored bounding boxes, along with confidence level. The ResNet-50 model gives accuracy of 98% with training testing ratio as 90:10.

Figure 8 shows the results of detection applied on a video feed by the ResNet-50 model with ratios 80:20. The accuracy provided by the ResNet-50 model in this instance with an 80:20 ratio is 99%. In this dataset, 80% of the data are training data, while 20% are testing data.

Figure 9 shows the results of person and face mask detection applied on a video feed by the ResNet-50 model with 70:30 train and test dataset ratio. The accuracy provided by the ResNet-50 model in this instance with a 70:30 ratio is 99%. In this dataset, 70% of the data are training data, while 30% are testing data.



Fig. 6 Frame in the considered input video

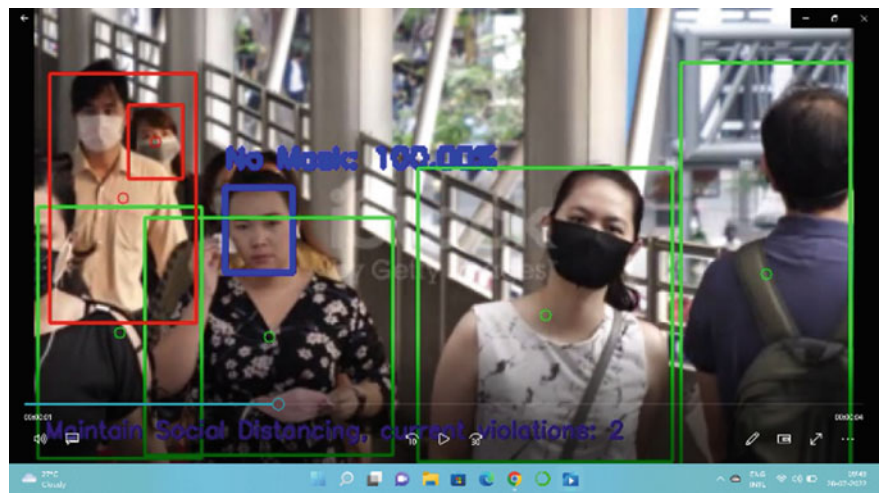


Fig. 7 Person and face mask detection result for ResNet-50 model with 90:10 train and test dataset ratios

Figure 10 shows the results of detection applied on a video feed by using the ResNet-50 model. People violating social distancing are shown with a red bounding box, whereas the ones following it are bounded by green boxes. The accuracy provided by the ResNet-50 model in this instance with an 60:40 ratio is 97%. In this dataset, 60% of the data are training data, while 40% are testing data. The model detects the face mask, and if a violation is found, it binds the person with a blue box and gives an accuracy that is shown in Fig. 10.

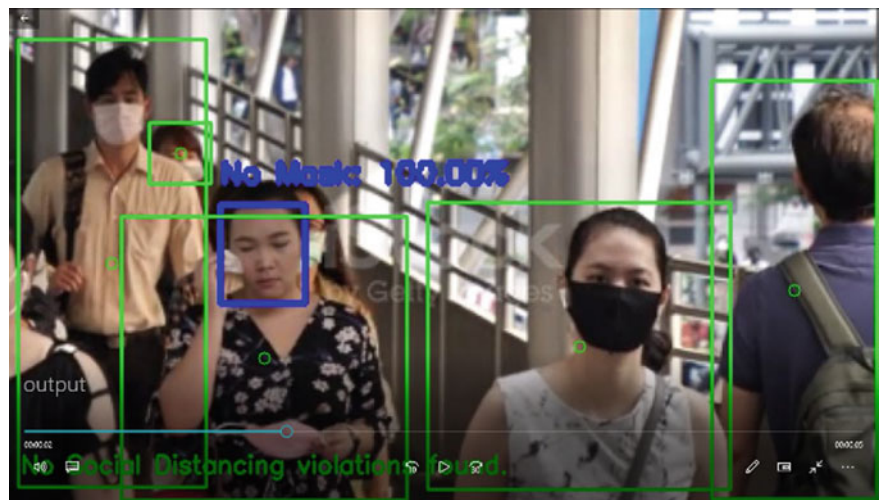


Fig. 8 Person and face mask detection result for ResNet-50 model with 80:20 train and test dataset ratios



Fig. 9 Person and face mask detection result for ResNet-50 model with 70:30 train and test dataset ratios

Figure 11 shows the YOLOv5 model social distancing result. It examines a video stream frame by frame and detects the social distance between people. If there is a violation, a box appears over the group of people with a message “high risk” in red color (as an indication of danger); otherwise “low risk” is displayed in yellow color (refer Fig. 10). YOLOv5 gives an accuracy of 55%. The YOLOv5 model failed to process the training and testing ratios of 90:10, 80:20, 70:30, 60:40, whereas the

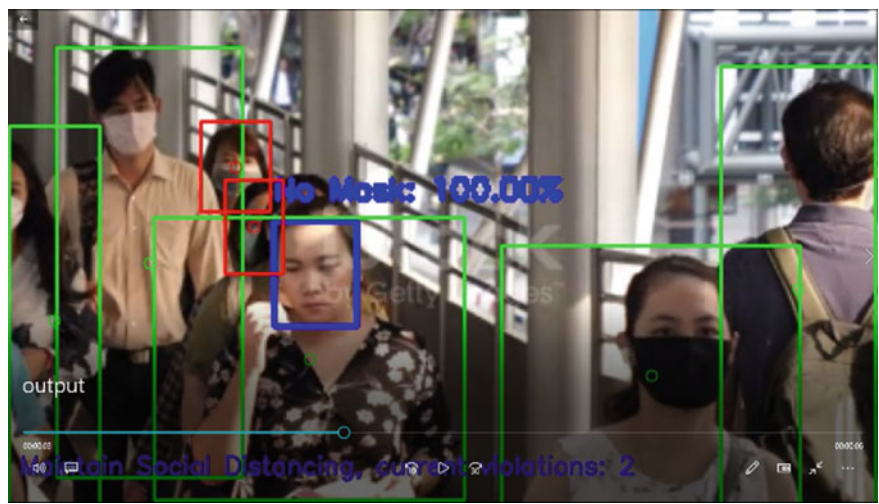


Fig. 10 Person and face mask detection for ResNet-50 model with 60:40 train and test dataset ratios

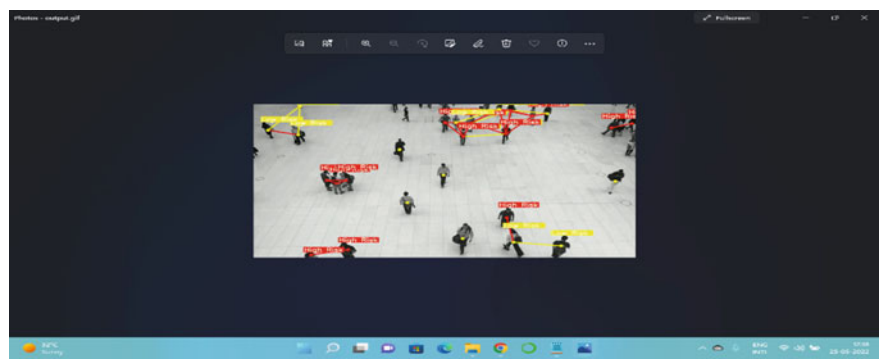


Fig. 11 Social distancing detection result for YOLOv5 model

ResNet-50 model resulted with an accuracy of 98% and YOLOv5 with much less accuracy than expected.

Figure 12 shows the results of detection applied on a video feed by the MobileNet model. The accuracy provided by the MobileNet model using a 90:10 ratio is 99% in this scenario, where the dataset is split into 90% training data and 10% testing data. MobileNet model gives the highest accuracy of 99% when compared with YOLOv5 (55%) and ResNet-50 (98%) with 90:10 dataset ratio.

Figure 13 shows the result of person and face mask detection applied on a video feed by the MobileNet model. The accuracy provided by the MobileNet model in this instance with an 80:20 ratio is 99%. In this dataset, 80% of the data are training data, while 20% are testing data. MobileNet model gives the highest accuracy of

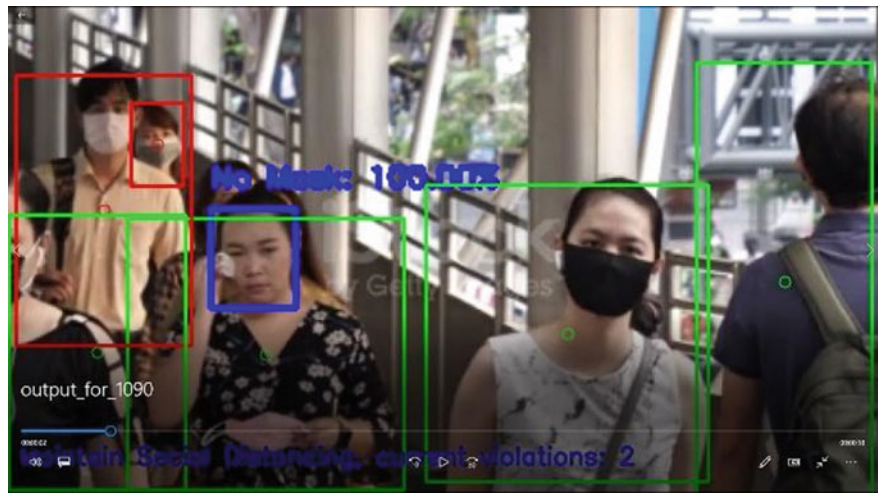


Fig. 12 Person and face detection result for MobileNet model with 90:10 train and test dataset ratios



Fig. 13 Person and face detection result for MobileNet model with 80:20 train and test dataset ratios

99% when compared with YOLOv5 (55%) and ResNet-50 (98%) with 80:20 dataset ratio.

Figure 14 shows the results of detection applied on a video feed by using the MobileNet model. People violating social distancing are shown with a red bounding box, whereas the one following it is bounded by green boxes. In this case, the model is with a dataset where 70% is training data and 30% is testing data, and the accuracy



Fig. 14 Person and face detection result for MobileNet model with 70:30 train and test dataset ratios



Fig. 15 Person and face detection result for MobileNet with 60:40 train and test dataset ratios

given by MobileNet model with 70:30 ratio is 99%. The model detects the face mask; if a violation is found, it binds the person with a blue box and gives an accuracy that is shown in Fig. 13. The MobileNet model has given the highest accuracy of 99% when compared with YOLOv5 (55%) and ResNet-50 (98%).

Figure 15 shows the results of detection applied on a video feed by the MobileNet model. In this case, the model is fed with a 60:40 dataset, where 60% is training data

and 40% is testing data. The accuracy obtained by the MobileNet model with 60:40 ratio is 99%. MobileNet model gives the highest accuracy of 98% when compared with YOLOv5 (55%) and ResNet-50 (98%).

5.1 Comparison of Results

When applied with, the MobileNet model results with an accuracy of 99.43% and ResNet-50 model with an accuracy of 98.41, when the division of the training and testing datasets are 90:10 ratio. When the division of the training and testing datasets is considered with 80:20 and 70:30 ratios, the results are similar in both cases giving an accuracy of 99.23% for MobileNet and 99.12% for ResNet-50. When the training and testing datasets are split into 60:40, there is a 1% reduction in accuracy which amounts to 98.22% for MobileNet and 97.99% for ResNet-50.

5.2 Confusion Matrix

The confusion matrix for the MobileNet Model and ResNet-50 Model is given in Table 2a, b, respectively. The confusion matrix in Table 2 gives the model’s classification accuracy using two classes: masked and unmasked. From Table 2a, it can be observed that 99% of entries are marked correctly, i.e., true positive for the masked label and 1% are marked false negative for the masked category. About 99% are marked true negative for the non-masked category and 1% were marked for the false positive. From Table 2b, it can be observed that 98% of entries are marked correctly, i.e., true positives for the masked label, and 2% are marked false negative for the masked category. About 97% are marked true negative for the non-masked category and 3% are marked for the false positive.

Table 2 a, b Confusion matrix for MobileNet and ResNet-50, respectively

(a)			
True label	Masked	0.99	0.01
	Non-masked	0.01	0.99
		Masked	Non-masked
		Predicted label	
(b)			
True label	Masked	0.98	0.02
	Non-masked	0.03	0.97
		Masked	Non-masked
		Predicted label	

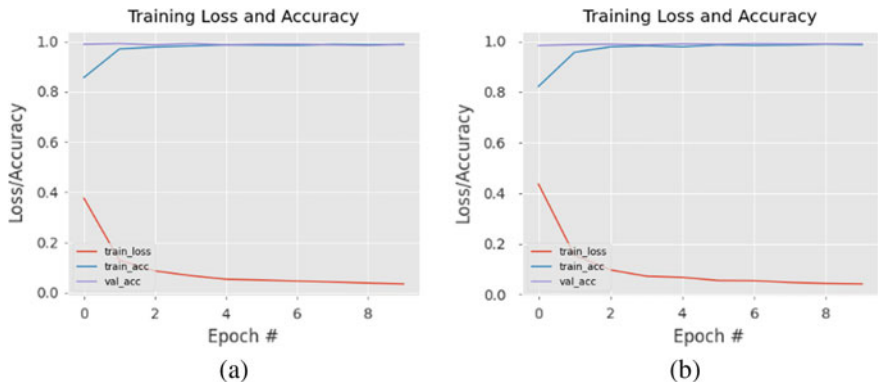


Fig. 16 a, b Training loss and accuracy for MobileNet with 90:10 train and test dataset ratios and with 80:20 train and test dataset ratios, respectively

As stated earlier, to train and test these models, 90:10; 80:20; 70:30; 60:40 training and testing data ratios are used to determine the model’s accuracy in this research. For all these dataset ratios, the MobileNet model gives accuracy between 98 and 99%, with batch size = 32, epochs = 10, and training hours greater than 1. Figure 15 shows the output for MobileNet when the training and testing data are considered with the ratio 90:10.

Figures 16 and 17 depict the graph between the loss/accuracy and the number of epochs for MobileNet, and Figs. 18 and 19 depict for ResNet for different training and testing ratios considered in this research. They (refer Figs. 16, 17, 18, and 19) depict how the increase in the epochs of training decreases the training loss. After the first epoch, it can be observed that the training accuracy has reached a maximum value of 1.0, whereas training loss is gradually approached 0.0. There is a sharp change in the accuracy and loss for training set after completion of first epoch (refer Figs. 16, 17, 18, and 19). Validation accuracy is consistently at 1.0 throughout the epoch progression.

Figures 16a, b show the output for MobileNet when the training and testing ratio datasets are considered with the ratio 90:10 and 80:20, respectively. Figure 17a shows the output for MobileNet when the training and testing datasets are considered with the ratio 70:30. Figure 17b shows the output for MobileNet when the training and testing data are considered with the ratio 60:40. The precision, recall, F1-score, and support metrics that are obtained for MobileNet are given in Table 3 for 90:10 training and testing dataset ratios.

Figure 18a shows the output obtained for ResNet when the training and testing data are considered with the ratio 90:10. Figure 18b shows the output for ResNet when the training and testing data are considered with the ratio 80:20.

Figure 19a shows the output for ResNet when the training and testing data are considered with the ratio 70:30. Figure 19b shows the output for ResNet when the training and testing data are in the ratio 60:40. Table 4 shows the precision, recall,

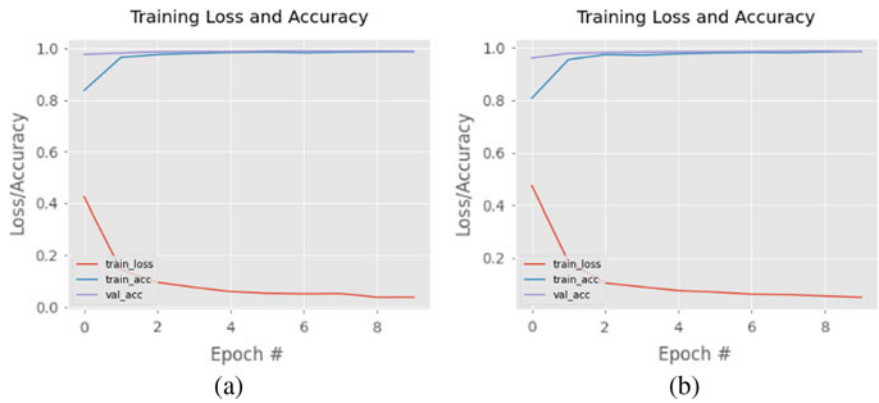


Fig. 17 a, b Training loss and accuracy for MobileNet with 70:30 train and test dataset ratios and with 60:40 train and test dataset ratios, respectively

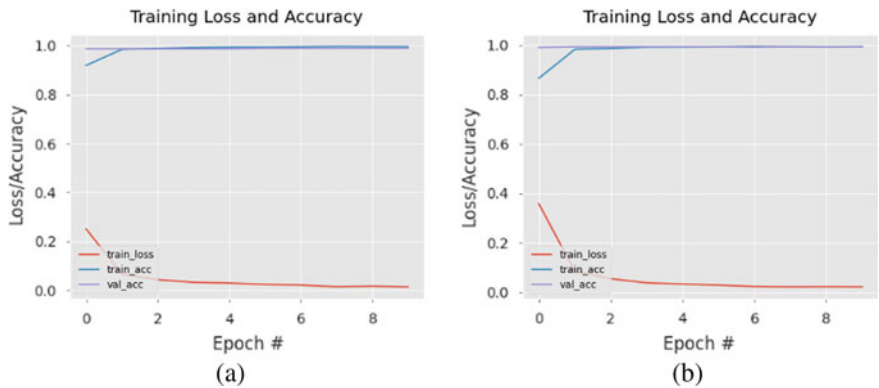


Fig. 18 a, b Training loss and accuracy for ResNet with 90:10 train and test dataset ratios and with 80:20 train and test dataset ratios, respectively

F1-score, and support values obtained using ResNet for 90:10 training and testing dataset ratios.

From the deep learning models implemented in this research, MobileNet model successfully processed the various ratios of training and testing data provided to it than the other models. Compared to YOLOv5 and ResNet-50, the MobileNet model obtained highest accuracy between 98 and 99% for the 90:10, 80:20, 70:30 and 60:40 training and testing dataset ratios. With 90:10, 80:20, and 70:30 ratios, MobileNet model obtains the accuracy of 99.43%. While with the 60:40 training and testing dataset ratio, there is a 1% reduction in accuracy, i.e., 98.22% for MobileNet and 97.99% for ResNet-50. Therefore, it can be noticed that MobileNet model is the best suitable model for face mask and social distance detection among the other models applied in this research. Comparison of various metrics' precision, recall, F1-score,

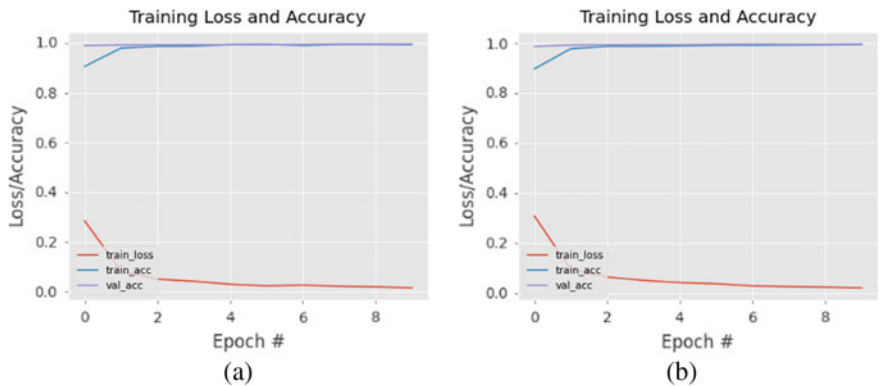


Fig. 19 a, b Training loss and accuracy for ResNet with 70:30 train and test dataset ratios and with 60:40 train and test dataset ratios, respectively

Table 3 Precision, recall, F1-score, and support metrics obtained for MobileNet model with 90:10 training and testing dataset ratios

	Precision	Recall	F1-score	Support
With mask	0.99	0.99	0.99	575
Without mask	0.99	0.99	0.99	575

Table 4 Precision, recall, F1-score, and support metrics obtained for ResNet model with 90:10 training and testing dataset ratios

	Precision	Recall	F1-score	Support
With mask	0.99	1.00	0.99	383
Without mask	1.00	0.99	0.99	384

and support calculated with and without mask for MobileNet and ResNet-50 models with 90:10 training and testing dataset ratios is depicted in Fig. 20.

6 Conclusion

This study provides an effective way for monitoring social distancing practices and in public places when manual monitoring is difficult. For the purpose of detecting people, identifying social distance, detecting faces, and categorizing face masks, the mostly used and efficient deep learning models in the literature such as MobileNet, YOLO, and ResNet-50 are applied, and results are compared in this research. This system determines if the two people are within 6 feet of the frame for social distance

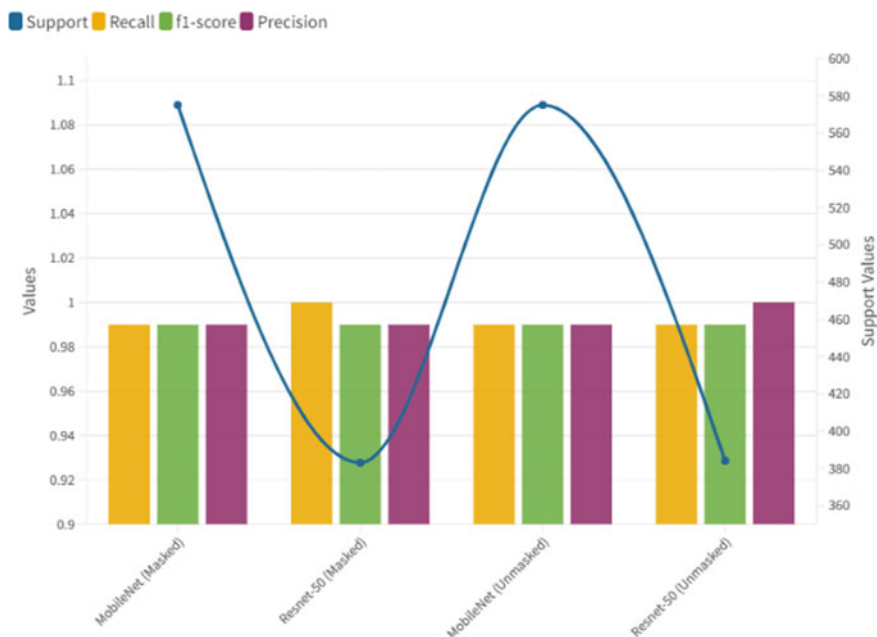


Fig. 20 Comparison of performance metrics for MobileNet and ResNet-50 models

by calculating the Euclidean/Manhattan distance; if not, a warning is issued. It additionally supplies strategies for boosting data to cope with blurry photographs, and films shot in low-light situations have been evaluated. The training and testing ratios of 90:10, 80:20, 70:30, and 60:40 are implemented in the Anaconda + Python 3.x (3.8 or older) environment, and the results for the deep learning models are compared. From the results, it can be observed that MobileNet gives the highest accuracy of 99% when considered with training and testing dataset ratios of 90:10 which gives better results compared to other models and other training and testing ratios. This implemented application in this research can be helpful in crowded places such as hospitals and clinics, airports, shops and workplaces, where social distancing and face masks are required to avert the spread of not only COVID-19 but also other communicable diseases as well.

References

1. Ahama, A.H., Zaini, N., Latip, M.F.A.: Person detection for social distancing and safety violation alert based on segmented ROI. Institute of Electrical and Electronics Engineers (IEEE) (2020)
2. Limbasiya, B., Raut, C.: COVID-19 face mask and social distancing detector using machine learning. Int. Res. J. Eng. Technol. (IRJET) (2021)

3. WHO Coronavirus (COVID-19) Dashboard. <https://covid19.who.int>. Accessed on 14 Aug. 2022
4. Symptoms of coronavirus. U.S. Centers for Disease Control and Prevention (CDC), 22 Feb. 2021. Archived from the original on 4 March 2021. Retrieved 4 March 2021. <https://www.cdc.gov/coronavirus/2019-ncov/symptoms-testing/symptoms.html>. Accessed on 14 Aug. 2022
5. Bhambani, K., Jain, T., Sultanpure, K.A.: Real-time face mask and social distancing violation detection system using You Only Look Once (YOLO). Institute of Electrical and Electronics Engineers (IEEE) (2020)
6. Singh, R.R., Nayak, N., Srinivasan, S., Biradar, R.R.: COVID-19 monitoring system using social distancing and face mask detection on surveillance video datasets (2021). arXiv
7. Walia, I.S., Kumar, D., Sharma, K., Hemanth, J.D., Popescu, D.E.: An integrated approach for monitoring social distancing and face mask detection using stacked ResNet-50 and YOLOv5. Multidisciplinary Digital Publishing Institute (MDPI) (2021)
8. Krishna, K.S.P., Harshita, S., Deep, N., Gautham, M.G., Deepthi, K.: Social distancing and face mask detection using deep learning. Int. Res. J. Eng. Technol. (IRJET) (2021)
9. Hou, Y.C., Baharuddin, M.Z., Yussof, S., Dzulkifly, S.: Social distance detection using DL model. Institute of Electrical and Electronics Engineers (IEEE) (2020)
10. More, A.S., Yendhe, S.K., Chandane, P.D., Sinalkar, B.D.: Detection of face mask and social distancing using ML algorithm. Int. J. Creative Res. Thoughts (IJCRT) (2022)
11. Hassan, F., Roy, S., Prabhakar, M., Rawat, H.: Social distance and face mask detection (2021)
12. Sridhar, S.B., Reddy, P., Pranay, V., Preetha, D.: Face mask and social distancing detector. Int. Res. J. Modern Eng. Technol. Sci. (IRJMETS) (2021)
13. Khan, M.I., Borika, B., Rai, G., Kharapkar, A., Likhar, C.: Review on real time face mask and social distancing detection system. Int. J. Sci. Res. Dev. (IJSRD) (2021)
14. Savita, S., Nikhil, D.K., Vishwesh, M.S.: Social distancing and face mask detection from CCTV camera (2021)
15. Gawde, B.B.: A fast, automatic risk detector for COVID-19. Institute of Electrical and Electronics Engineers (IEEE) (2020)
16. Kumar, G.S., Shetty, S.D.: Application development for mask detection and social distancing violation detection using convolutional neural networks (2021)
17. Sathyabama, B., Devpura, A., Maroti, M., Rajput, R.S.: Monitoring pandemic precautionary protocols using real-time surveillance and artificial intelligence. Institute of Electrical and Electronics Engineers (IEEE) (2020)
18. Kaviya, V., Vedhashree, D., Rani, T.P.: A novel Covid prevention method using deep learning. Institute of Electrical and Electronics Engineers (IEEE) (2021)
19. Boulila, W., Alzahem, A., Almoudi, A., Afifi, M., Alturki, I., Driss, M.: A deep learning-based approach for real-time facemask detection (2021). arxiv
20. COCO dataset news. <https://cocodataset.org>. Accessed on 5 May 2022
21. Tensorflow WIDER. FACE dataset. https://www.tensorflow.org/datasets/catalog/wider_face. Accessed on 15 May 2022
22. GitHub X-s RMFD. <https://github.com/X-zhangyang/Real-World-Masked-Face-Dataset>; <https://www.v7labs.com/open-datasets/rmfd>. Accessed on 17 May 2022
23. COCO-stuff category. <https://arxiv.org/abs/1612.03716>. Accessed on 27 March 2022